PHYSICS

1)

List-I		List-II	
(I)	$\frac{d}{dx}(x\ell nx)$	(P)	$\frac{1}{(2x+1)^2}$
(II)	$\frac{d}{dx}\left(\frac{x}{2x+1}\right)$	(Q)	12
(III)	$\frac{d^2}{dx^2}\left(6x^2\right)$	(R)	1 + □nx
(IV)	$\frac{d}{dx}\left(\frac{1}{2x+1}\right)$	(S)	$-\frac{2}{(2x+1)^2}$

(1) I → R;II → P;III → Q;IV → S

(2) I → S;II → R;III → Q;IV → P

(3) I → P;II → Q;III → R;IV → S

(4) I → Q;II → S;III → P;IV → R

2) A Body moves 6 m north, 8 m east and 10m vertically upwards, what is its resultant displacement from initial position:-

(1) $10\sqrt{2}$m

(2) 10m

(3) $\frac{10}{\sqrt{2}}$m

(4) 10 × 2m

3)

If $\tan\theta = \frac{1}{\sqrt{5}}$ and θ lies in the first quadrant, the value of cos θ is :

(1) $\sqrt{\frac{5}{6}}$

(2) $-\sqrt{\frac{5}{6}}$

(3) $\frac{1}{\sqrt{6}}$

(4) $\frac{1}{-\sqrt{6}}$

4) $\int_{-1}^{+1} t^{-3} dt$ = ?

(1) 0
(2) 1
(3) 2
(4) –2

5)

cos 425°=?

(1) cos 65º
(2) cos 125º
(3) sin 25º
(4) sin 65º

6) What is the value of cosec 330°

(1) –2
(2) $-\frac{1}{2}$
(3) 2
(4) $+\frac{1}{2}$

7) $\frac{d}{dx}(x^2e^x) =$

(1) $x^2e^x + e^xx^2$
(2) $x^2e^x + e^x2x$
(3) e^x2x
(4) $x^2e^x - e^x2x$

8) Calculate distance between point A (6, 7, 2) and point B (9, 12, 5)

(1) $\sqrt{45}$ unit
(2) $\sqrt{43}$ unit
(3) $\sqrt{44}$ unit
(4) $\sqrt{42}$ unit

9) Value of sin(45º + θ) cos(15º +θ)
– cos(45º + θ)sin(15º + θ) is :-

(1) 1

(2) $\frac{3}{2}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

10) If sin 300°=a and cos 120° = b, then

(1) a > 0, b > 0

(2) a > 0, b < 0

(3) a < 0, b > 0

(4) a < 0, b < 0

11) Radius of a sphere is increasing at rate of 3 m/s then find rate of change of its volume when its radius is 1 m.

(1) $36\pi\frac{m^3}{sec}$

(2) $12\pi\frac{m^3}{sec}$

(3) $9\pi\frac{m^3}{sec}$

(4) $27\frac{m^3}{sec}$

12) If $y = 2\sin^2\theta + \tan\theta$ then $\frac{dy}{d\theta}$ will be

(1) $4\sin\theta\cos\theta + \sec\theta\tan\theta$

(2) $2\sin2\theta + \sec^2\theta$

(3) $4\sin\theta + \sec^2\theta$

(4) $2\cos2\theta + \sec^2\theta$

13) $y = \frac{1}{x+1}$, find $\frac{dy}{dx}$

(1) $-\frac{2}{(x+1)^2}$

(2) $-\frac{1}{(x-1)^2}$

(3) $-\frac{1}{(x+1)^2}$

(4) $\frac{1}{(x+1)^2}$

14) Evaluate: $\int \left(x^2 - \cos x + \frac{1}{x}\right) dx$

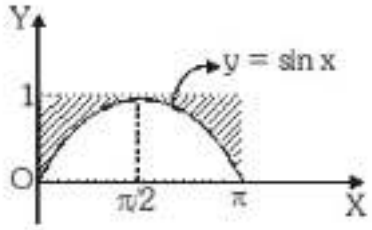
(1) $\frac{x^3}{3} - \sin x + \log_{10} x + c$

(2) $\frac{x^2}{3} - \sin x + \log_e x + c$

(3) $\frac{x^3}{3} - \sin x + \log_e x + c$

(4) $\frac{x^2}{3} + \sin x + \log_e x + c$

15) Graph of y = sin x is given. Area of shaded region is:



(1) $\pi - 2$

(2) 2

(3) $\pi + 2$

(4) π

16) A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be

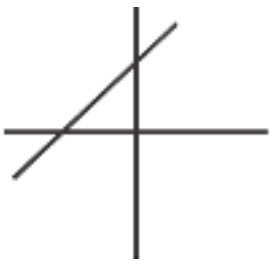
(1) 0

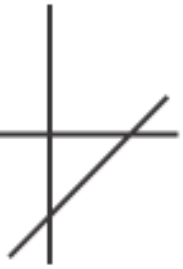
(2) 3

(3) 12

(4) 47

17) The equation of a straight line is 2x – 3y = 5 . Its graph is

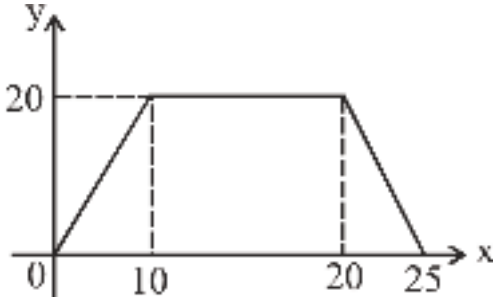
(1) 

(2) 

(3) 

(4) 

18) Calculate = $\int_0^{25} y \, dx$?



(1) 150

(2) 250

(3) 350

(4) 450

19) Match the given columns.

Column-I		Column-II	
(P)	$\frac{d}{dx}[x^2 + 3x + 4]_{\text{at } x = 0}$	(A)	$\frac{1}{3}$
(Q)	$\frac{d}{dx}[3\sin x - 4\cos x]_{\text{at } x = \frac{\pi}{2}}$	(B)	4

(R)	$\int_{x=0}^{x=1}(x^2-2x+1)dx$	(C)	3
(S)	$\int_{-\infty}^{0} e^x dx$	(D)	1

(1) P-B,Q-C,R-B,S-A

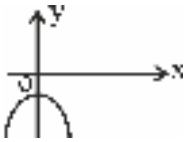
(2) P-A,Q-B,R-C,S-D

(3) P-C,Q-B,R-A,S-D

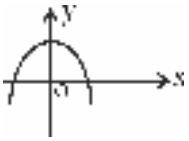
(4) P-D,Q-C,R-B,S-D

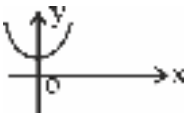
20)

$y = -3x^2 - 5$, then graph can be :

(1) 

(2)

(3) 

(4) 

21) Find the minimum value of x, when
$x = 3\sin\theta - 4\cos\theta + 5$:-

(1) Zero

(2) 5

(3) –5

(4) 10

22) $1 + 3 + 5 + 7 + \ldots + 33 = ?$

(1) 289

(2) 298

(3) 189

(4) 198

23) The maximum and minimum magnitude of resultant of two given vectors are 17 units and 7 units respectively. If these two vectors are at right angle to each other, then magnitude of their resultant is :-

(1) 18

(2) 16

(3) 13

(4) 14

24) Find component of $3\hat{i}+4\hat{j}+\hat{k}$ along $\hat{i}+\hat{j}+\hat{k}$

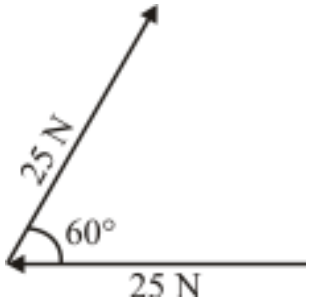
(1) $\frac{8}{3}(\hat{i}+\hat{j}+\hat{k})$

(2) $\frac{8}{3}(\hat{i}-\hat{j}+\hat{k})$

(3) $\frac{8}{3}(\hat{i}-\hat{j}-\hat{k})$

(4) $\frac{8}{3}(3\hat{i}+4\hat{j}+\hat{k})$

25) Two forces each of the magnitude 25 N are applied on a point as shown. Their resultant force is :-



(1) $25\sqrt{3}\,\text{N}$

(2) 25N

(3) 50N

(4) 12.5N

26) If $\left|\hat{A}+\hat{B}\right|=\sqrt{3}$ then find $\left(2\hat{A}+3\hat{B}\right)\cdot\left(\hat{A}+3\hat{B}\right)$:-

(1) 15

(2) $\frac{10\sqrt{3}}{2}$

(3) $\frac{31}{2}$

(4) 10

27) The unit vector parallel to the resultant of the vectors $\vec{A}=4\hat{i}+3\hat{j}+6\hat{k}$ and $\vec{B}=-\hat{i}+3\hat{j}-8\hat{k}$ is :

(1) $\frac{1}{7}(3\hat{i}+6\hat{j}-2\hat{k})$

(2) $\frac{1}{7}(3\hat{i}+6\hat{j}+2\hat{k})$

(3) $\frac{1}{49}(3\hat{i}+6\hat{j}-2\hat{k})$

(4) $\frac{1}{49}(3\hat{i}-6\hat{j}+2\hat{k})$

28) Which of the sets given below may represent. The magnitude of three vectors adding to zero ?

(1) 2,4,8
(2) 4,8,16
(3) 1,2,1
(4) 0.5,1,2

29) The angle between two vectors given by $6\hat{i}+6\hat{j}-3\hat{k}$ and $7\hat{i}+4\hat{j}+4\hat{k}$ is :-

(1) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(2) $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$
(3) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$
(4) $\sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

30)

If the vectors $\vec{A}=3\hat{i}+\hat{j}-3\alpha\hat{k}$ and $\vec{B}=\hat{i}+2\alpha\hat{j}-2\hat{k}$ are perpendicular to each other, then the value of α is :

(1) $-\frac{3}{8}$
(2) $-\frac{1}{8}$
(3) $-\frac{2}{3}$
(4) 8

31) If $\vec{A}\times\vec{B}=\vec{R}$, then incorrect statement is :-

(1) $\vec{R}\perp\vec{A}$
(2) $\vec{R}\perp\vec{B}$
(3) $\vec{R}\perp(\vec{A}+\vec{B})$
(4) $\vec{R}\perp(\vec{A}\times\vec{B})$

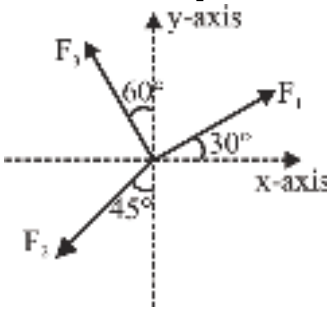
32) If $\left|\vec{a}\times\vec{b}\right| = \frac{1}{\sqrt{3}}\left(\vec{a}\cdot\vec{b}\right)$, then $\left|\vec{a}+\vec{b}\right|$ is :-

(1) $\sqrt{a^2+b^2+ab}$
(2) $\sqrt{a^2+b^2+2ab}$

(3) $\sqrt{a^2 + b^2}$

(4) $\sqrt{a^2 + b^2 + \sqrt{3}ab}$

33) If three coplanar forces acting at origin are in equillibrium then which is correct :-



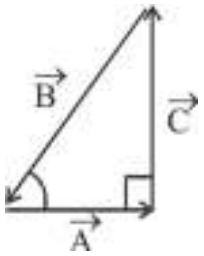
(1) $F_1 - F_2 = \frac{F_3}{\sqrt{2}}$

(2) $F_1 - F_2 = \sqrt{\frac{2}{3}}F_3$

(3) $F_1 + F_3 = \sqrt{2}F_2$

(4) $F_1 - F_3 = \frac{2}{\sqrt{3}}F_2$

34) In the figure what is the angle between $\vec{A}$ and $\vec{B}\left(\text{given } C = \frac{B}{2}\right)$:-



(1) 30°

(2) 60°

(3) 150°

(4) 120°

35)

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 4\hat{k}$, then find out $\left|3\vec{a} + \vec{b}\right|$.

(1) $3\sqrt{10}$

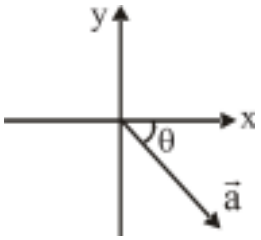
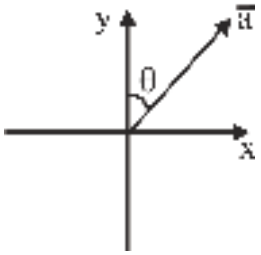
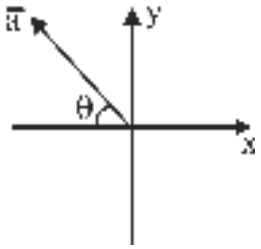
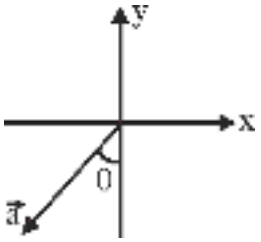
(2) 10

(3) $10\sqrt{3}$

(4) 16

36) Match vector $\vec{a}$ shown in column-I with its unit vector representation in column-II

	Column-I		**Column-II**

(A)		(P)	$\vec{a} = a\sin\theta\hat{i} + a\cos\theta\hat{j}$
(B)		(Q)	$\vec{a} = -a\cos\theta\hat{i} + a\sin\theta\hat{j}$
(C)		(R)	$\vec{a} = -a\sin\theta\hat{i} - a\cos\theta\hat{j}$
(D)		(S)	$\vec{a} = a\cos\theta\hat{i} - a\sin\theta\hat{j}$

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-R, C-P, D-S

(3) A-S, B-P, C-Q, D-R

(4) A-R, B-Q, C-P, D-S

37) The displacement of the tip of the 10 cm long minute hand of a clock between 12 : 15 pm to 12 : 45 pm is :-

(1) 10 cm

(2) 20 cm

(3) $10\sqrt{2}$ cm

(4) 15 cm

38) A boy moves along a curved path of a semicircle of radius 50 m. The ratio of distance covered to the displacement is

(1) $\frac{2}{\pi}$

(2) 1 : 1

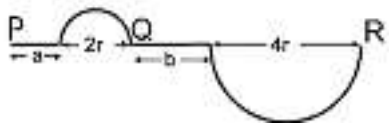
(3) π

(4) $\frac{\pi}{2}$

39) A drunkard walking in a narrow lane takes 5 steps forward and 2 steps backward followed again by 5 steps forward and 2 steps backward and so on. Each step is 1 m long and requires 1 s. The time the drunkard takes to fall in a pit 13 m away is

(1) 25 s
(2) 32 s
(3) 37 s
(4) 49 s

40) A car starts from P and follows the path as shown in figure. Finally car stops at R. If a = 7 m, b = 8 m and $r = \frac{11}{\pi}m$? [Take $\pi = \frac{22}{7}$]. The distance travelled by the car is



(1) 36 m
(2) 38 m
(3) 46 m
(4) 48 m

41) A particle moves along a circular arc of radius R making an angle of θ at centre. The magnitude of displacement is

(1) $2R\sin\theta/2$
(2) $2R\sin\theta$
(3) $R\sin\theta/2$
(4) $R\sin\theta$

42) A person moves 20 m towards north then 30 m towards east and finally $40\sqrt{2}\,m$ south-west. His displacement is

(1) $10\sqrt{5}\,m$, $\tan^{-1}$ (2) S of W
(2) $20\sqrt{5}\,m$, $\tan^{-1}$ (2) S of W
(3) $20\,m$, S – W
(4) $10\sqrt{5}\,m$, S – W

43) If a particle moves from point P(2, 3, 5) to point Q (3, 4, 5). Its displacement vector be:-

(1) $\hat{i}+\hat{j}+10\hat{k}$
(2) $\hat{i}+\hat{j}+5\hat{k}$
(3) $\hat{i}+\hat{j}$
(4) $2\hat{i}+4\hat{j}+6\hat{k}$

44) A person moves on a circular path of radius 5 m. What will be his displacement and distance after he turns 90°?

(1) $5\sqrt{2}\,m, 5\pi\,m$

(2) $5\,m, 2\sqrt{5}\,m$

(3) $5\,mt, 10\,\pi m$

(4) $5\sqrt{2}\,m, 2.5\,\pi m$

45) An athlete completes one round of a circular track of radius R in 40 sec. What will be his displacement at the end of 2 min. 20 sec

(1) Zero

(2) 2R

(3) $2\pi R$

(4) $7\pi R$

CHEMISTRY

1) A subshell in which n = 4 and = 2 is expressed by :-

(1) 4s

(2) 4p

(3) 4d

(4) 5p

2) The energy of a subshell in a hydrogen atom depends on:

(1) The value of **n**

(2) The value of **(n +)**

(3) Only on the value of

(4) The values of **n**, , and **m**

3) Which of the following set of quantum number is not valid :-

(1) n = 1, = 2

(2) n = 2, m = 1

(3) n = 3, = 0

(4) n = 4, = 2

4) Which of the following orbital is spherical?

(1) s

(2) p

(3) d

(4) f

5) Match **List-I** with **List-II** :

	List-I (quantum number)		List-II (Orbital)
(a)	n = 2, □ = 1	(i)	2 s
(b)	n = 3, □ = 2	(ii)	3 s
(c)	n = 3, □ = 0	(iii)	2 p
(d)	n = 2, □ = 0	(iv)	3 d

Choose the **correct answer** from the options given below :

(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

(3) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(4) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

6) Maximum possible electrons in a given shell is:

(1) n^2

(2) 4l + 2

(3) $2n^2$

(4) 2l + 1

7) Match the list-I with list-II:

List-I		List-II	
(a)	Size (radius) of orbit	(p)	Azimuthal quantum number(□)
(b)	Shape of orbital	(q)	Principal quantum number(n)
(c)	Orientation of orbital	(r)	Magnetic quantum number(m)
(d)	Spin of electron	(s)	Spin quantum number(s)

(1) (a – q), (b – p), (c – s), (d – r)

(2) (a – q), (b – p), (c – r), (d – s)

(3) (a – r), (b – s), (c – p), (d – q)

(4) (a – q), (b – r), (c – p), (d – s)

8) For electrons in ' 2 s ' and ' 2 p ' orbitals, the orbital angular momentum values, respectively are :

(1) 0 and $\sqrt{6}\frac{h}{2\pi}$

(2) 0 and $\sqrt{2}\frac{h}{2\pi}$

(3) $\sqrt{2}\frac{h}{2\pi}$ and 0

(4) $\frac{h}{2\pi}$ and $\sqrt{2}\frac{h}{2\pi}$

9) The portion of orbital diagrams representing the electrons configuration of certain element shown below. Which of these violate Pauli’s Exclusion principle ?

(1) 

(2) 

(3) 

(4) 

(1) Only 1

(2) Only 3

(3) 2 and 4

(4) 1, 2 and 4

10) The electrons identified by quantum numbers n and ℓ :-
(a) n = 4, ℓ = 1
(b) n = 4, ℓ = 0
(c) n = 3, ℓ = 2
(d) n = 3, ℓ = 1
Can be placed in order of increasing energy as

(1) (a) < (c) < (b) < (d)

(2) (c) < (d) < (b) < (a)

(3) (d) < (b) < (c) < (a)

(4) (b) < (d) < (a) < (c)

11) The total values of 'm' for the electrons in n = 3 is:-

(1) 3

(2) 8

(3) 9

(4) 4

12) No. of all subshells of n + □ = 7 is:-

(1) 4

(2) 5

(3) 6

(4) 7

13) Consider the given sets of quantum numbers
(a) n = 4,□ = 2, m = –2, s = $+\frac{1}{2}$
(b) n = 3, □ = 3, m –3, s = $\frac{-1}{2}$
(c) n = 2, □ = 1, m = +1, s = +1

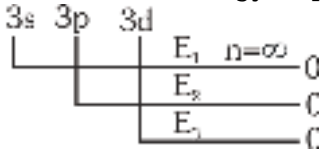
(d) n = 1, ℓ = 0, m = 0, s = 0
Choose the option containing the sets which are not possible.

(1) (a), (b) and (c) only
(2) (c) and (d) only
(3) (b), (c) and (d) only
(4) (b) and (d) only

14) Maximum number of electrons in a subshell of an atom is determined by the following :-

(1) $2n^2$
(2) $4\ell + 2$
(3) $2\ell + 1$
(4) $4\ell - 2$

15) For H atom, the energy required for the removal of electron from various sub-shells is given as under:-



The order of the energies would be :-

(1) $E_1 > E_2 > E_3$
(2) $E_3 > E_2 > E_1$
(3) $E_1 = E_2 = E_3$
(4) None of these

16)

Any nf–orbital can accomodate upto :–

(1) 14 electron
(2) Six electrons
(3) Two electrons with parallel spin
(4) Two electrons with opposite spin

17) n, ℓ and m values of an electron in $3p_y$ orbital are :-

(1) n = 3 ; ℓ = 1 and m = 1
(2) n = 3 ; ℓ = 1 and m = –1
(3) Both 1 and 2 are correct
(4) None of these

18) Which orbital has two angular nodes?

(1) s
(2) p
(3) d

(4) f

19) What is the maximum number of electrons that can be associated with the following set of quantum numbers?
$n = 3,\ \ell = 1$ and $m = -1$

(1) 2

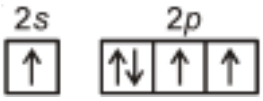
(2) 10

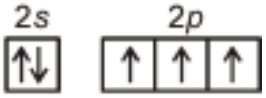
(3) 6

(4) 4

20) The orbital diagram in which Aufbau principle is violated is

(1) 

(2) 

(3) 

(4) 

21) **Assertion (A) :** For Hydrogen like species, energy follow the order 3s = 3p = 3d
Reason (R) : Energy of H-like species depends on Principal quantum number (n) only.

(1) Both A and R are correct

(2) Both A and R are incorrect

(3) Only A is correct

(4) Only R is correct

22) **Statement-I:** Half filled, full filled degenerate orbitals are more stable.
Statement-II: Extra stability is due to the symmetrical distribution of electrons and exchange energy.

(1) Both the Statements are correct

(2) Statement-I is correct, But Statement-II is wrong

(3) Statement-I is wrong, But Statement-II is correct

(4) Both the Statements are incorrect

23) In Cr atom number of electrons with n = 3 are:-

(1) 13

(2) 12

(3) 14

(4) 10

24) Which one is a wrong statement ?

(1) The uncertainty principle is $\Delta E \times \Delta t \geq h/4\pi$

(2) Half filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.

(3) The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms

(4) de-Broglie wavelength is given by $\lambda = \frac{h}{m\nu}$, where m = mass of the particle, ν = velocity of the particle

25) Heisenberg Uncertainity principle is not valid for

(1) Moving electron

(2) Motor car

(3) Stationary particles

(4) 2 & 3 both

26)

	List-I		**List-II**
(A)	No of electrons present in an orbit	(1)	2
(B)	Number of orbitals in an orbit	(2)	n
(C)	Number of electrons in an orbital	(3)	n^2
(D)	Number of sub shells in an orbit	(4)	$2n^2$

The correct match is

(1) A → 4, B → 2, C → 1, D → 3

(2) A → 1, B → 2, C → 3, D → 4

(3) A → 4, B → 3, C → 1, D → 2

(4) A → 2, B → 1, C → 3, D → 4

27)

Spin angular momentum for electron will be :-

(1) $\frac{\sqrt{s(s+1)}}{\frac{h}{2\pi}}$

(2) $\frac{\sqrt{2s(s+1)}}{\frac{h}{2\pi}}$

(3) $\frac{\sqrt{s(s+2)}}{\frac{h}{2\pi}}$

(4) None

28) Magnetic moment 2.83 BM is given by which of the following ions ?

(Atomic Number Ti = 22, Cr = 24, Mn = 25, Ni = 28) :-

(1) Ti^{3+}

(2) Ni^{2+}

(3) Cr^{3+}

(4) Mn^{2+}

29) The correct relation according to Heisenberg's uncertainty principle is/are :

(1) $\Delta x \times \Delta p \geqslant \frac{h}{4\pi}$

(2) $\Delta E \times \Delta t \geqslant \frac{h}{4\pi}$

(3) $\Delta x \times \Delta v \geqslant \frac{h}{4\pi m}$

(4) All of the above

30) Given below are two statements:
Statement I : The energy of 3d orbital is greater than 4p orbital.
Statement II : If two orbital have the same value of (n + □), the ortbial with lower value of n will have the lower energy.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct.

(4) Both statements I and statements II are incorrect.

31) The uncertainty in momentum of an electron is 1×10^{-5} kg m/s. The uncertainty in its position will be: ($h = 6.62 \times 10^{-34}$ kg-m^2/s)

(1) 1.05×10^{-28} m

(2) 1.05×10^{-26} m

(3) 5.27×10^{-30} m

(4) 5.25×10^{-28} m

32) Number of waves in third orbit :

(1) 3

(2) 4

(3) 5

(4) 0

33) A pure substance can only be :

(1) A compound

(2) An element
(3) An element or a compound
(4) A heterogenous mixture

34) Which of the following is correct relation between g & amu ?

(1) 1 amu × N_A = 1g
(2) 1 amu = N_A × 1g
(3) 1 amu = 1g
(4) 1 amu = $\frac{1}{1g}$

35) Calculate the volume at STP occupied by 240 g of SO_2.

(1) 64 L
(2) 84 L
(3) 59 L
(4) 73 L

36)

Which has the maximum number of molecules among the following ?

(1) 44 g of CO_2
(2) 48 g O_2
(3) 8 g H_2
(4) 64 g SO_2

37)

The prefix pico stands for :-

(1) 10^{15}
(2) 10^{-15}
(3) 10^{-12}
(4) 10^{12}

38)

SI unit of the quantity electric current is

(1) Volt
(2) Ampere
(3) Candela
(4) Newton

39) 2.5 mol of calcium has mass :-

(1) 100 g
(2) 10g
(3) 5g
(4) 1g

40) 1 amu is equal to :-

(1) 1.00758 g
(2) 0.000549 g
(3) 1.67×10^{-24} g
(4) 6.02×10^{-23} g

41) One litre of a gas at STP weighs 1.25 g. It can possibly be :-

(1) C_2H_2
(2) CO
(3) O_2
(4) CH_4

42) Convert 3800 mm of Hg pressure into atm. pressure

(1) 1.5 atm
(2) 5 atm
(3) 3 atm
(4) 7.5 atm

43) Which statement is correct ?

(1) The properties of compound are altogether different from those of its constituent elements
(2) The SI system has seven base unit
(3) The SI unit of volume is litre
(4) Both (1) and (2)

44) Equal masses of O_2, H_2 and CH_4 are taken in a container. The respective mole ratio of these gases in container is :-

(1) 1 : 16 : 2
(2) 16 : 1 : 2
(3) 1 : 2 : 16
(4) 16 : 2 : 1

45) Actual mass of H_2O molecule :-

(1) 3×10^{-24} g

(2) 3×10^{-23} g

(3) 3×10^{-22} g

(4) 10^{-23} g

BIOLOGY

1) **Assertion :** Spiny skinned animals like star fish are called echinodermata.
Reason : These animals have an endoskeleton of calcareous ossicles.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

2) Which one of the following statements is incorrect about the occurrence of notochord ?

(1) In ascidian, it is present only in larval tail.

(2) It is replaced by a vertebral column in adult frog.

(3) It is absent through out life in human from the very beginning.

(4) It is present through out life in amphioxus.

3) **Statement-I :** Presense of mammary gland is a most unique feature of mammals.
Statement-II : In mammals, limbs help in walking, running, Climbing, burrowing, swimming or flying.

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect

(3) Only statement-I is correct.

(4) Only statement-II is correct.

4) The scales and paired fins are absent in

(1) Chondrichthyes

(2) Cyclostomes

(3) Osteichthyes

(4) All the fishes

5) Choose the option with same analogue
Echinodermata's water vascular system: locomotion, capture and transport of food and respiration**::**_____.

(1) Mollusc’s feather gills: excretion and respiration.

(2) Annelids' setae: movement and excretion.

(3) Arthropods' exoskeleton: respiration and protection.

(4) Chordates' notochord: support and movement.

6) Which of the following is a living fossil?

(1) Balanoglossus
(2) Echinus
(3) Ancylostoma
(4) *Limulus*

7) Which of the following pairs of animals comprises 'jawless fishes'?

(1) Mackerals and rohu
(2) Lampreys and hag fishes
(3) Guppies and hag fishes
(4) Lampreys and eels

8)

Which of the following statements (i – v) are incorrect?
(i) Parapodia are lateral appendages in arthropods used for swimming.
(ii) Radula in molluscs are structures involved in excretion.
(iii) Aschelminthes are dioecious.
(iv) Echinoderm adults show radial symmetry.
(v) Ctenophorans are diploblastic.

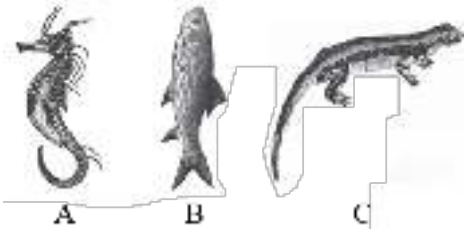
(1) (i) and (ii)
(2) (i) and (iii)
(3) (i), (iv) and (v)
(4) (iii) and (v)

9) Match the terms/feature given in column-I with their examples given in column-II and select the correct match from the option given below.

	Column-I (Term/Feature)		**Column-II (Examples)**
A.	Gregarious pest	(i)	Hirudinaria
B.	Vector	(ii)	Plannaria
C.	Oviparous with indirect development	(iii)	Sepia
D.	Metameres	(iv)	*Aedes*
E.	High regeneration capacity	(v)	*Locust*

(1) (i) (ii) (iii) (iv) (v)
(2) (iii) (v) (ii) (iv) (i)
(3) (iii) (i) (v) (ii) (iv)
(4) (v) (iv) (iii) (i) (ii)

10) Refer the figures A, B and C and choose the correct option which shows animals that regulate

buoyancy with the help of air bladder. 

(1) A and B
(2) A and C
(3) B and C
(4) All of the above

11) Which of the following sets of animals give birth to young ones?

(1) Platypus, Penguin, Bat, Hippopotamus
(2) Shrew, Bat, Cat, Kiwi
(3) Kangaroo, Hedgehog, Dolphin, Loris
(4) Lion, Bat, Whale, Ostrich

12) **A→** Chordata → Vertebrata → Cartilaginous skeleton → Placoid scales → No operculum
in This flow chart A is refers to:

(1) Osteichthyes
(2) Chondrichthyes
(3) Cyclostomata
(4) Amphibia

13) In which of the following phylum larvae are bilaterally symmetrical and adult are radically symmetrical ?

(1) Mollusca
(2) Echinodermata
(3) Annelida
(4) Arthropoda

14) Identify this animal, belongs to : 

(1) Hemichordata
(2) Urochordata

(3) Cephalochordata

(4) Vertebrata

15) Read the following statements A-D :-
(A) Members are worm like and marine
(B) Body divided into proboscis, collar and trunk
(C) Respiration by gills
(D) Flame cells are excretory organs
How many statements are incorrect for Balanoglossus.

(1) One

(2) Two

(3) Three

(4) Four

16) How many fishes in list given below have bony endoskelton :-
Scoliodon, Exocoetus, Pristis, Hippocampus, Labeo, Carcharodon, Clarias, Trygon, Pterophyllum, Catla.

(1) Five

(2) Six

(3) Seven

(4) Eight

17) In amphibian cloaca is common chamber for :-

(1) Alimentary canal + urinary tract

(2) Alimentary canal + Reproductive tract

(3) Alimentary canal + Respiratory tract

(4) Alimentary canal + urinary tract + Reproductive tract

18) (A) Bilateral symmetry
(B) Radial symmetry
(C) Diploblastic
(D) Triploblastic
(E) Acoelomate
(F) Round body on transverse section
Which of above given characters are present in Ascaris?

(1) A, D, F

(2) A, D, E, F

(3) B, C, E

(4) B, D, F

19) Which of the following is correct with respect to class, its character and example.

(1)	Aves – Forelimbs are modified into wings.	
(2)	Reptilia – Tympanum represents ear.	
(3)	Mammalia – Air sacs connected to lungs	
(4)	Amphibia – Homeothermic	

(1) 1

(2) 2

(3) 3

(4) 4

20) Intercostal muscles are found attached with :-

(1) Diaphragm

(2) Ribs

(3) Pleura

(4) Lungs

21) Maximum expiration after deep inspiration is called

(1) Vital capacity

(2) Lung capacity

(3) Tidal volume

(4) Residual volume

22) Arrange the following in an ascending order of volume –
(a) FRC (b) EC (c) RV (d) ERV

(1) d < c < b < a

(2) d < b < c < a

(3) d < a < b < c

(4) d < a < c < b

23) Which of the following statements is/are correct ?
(A) 97% of CO_2 is transported by RBCs in the blood.
(B) 20-25% CO_2 is transported by RBCs
(C) 70% of O_2 is carried as bicarbonate
(D) 7% of CO_2 is carried in a dissolved state through plasma

(1) A, B, C

(2) A, C

(3) B, D

(4) B, C, D

24) Sequence during inspiration:
1. Diaphragm contracts.
2. Thoracic cavity volume increases.
3. Intra-pulmonary pressure drops.
4. Air enters lungs.

(1) 1 → 2 → 3 → 4

(2) 2 → 1 → 3 → 4

(3) 1 → 3 → 2 → 4

(4) 1 → 2 → 4 → 3

25) Which statement is correct.
Statement-I : We have the ability to increase the strength of inspiration and expiration with the help of additional muscles.
Statement-II : Expiration takes place when the intra-pulmonary pressure is higher than the atmospheric pressure.

(1) Both Statements correct.

(2) Both Statements incorrect.

(3) Only Statement-I correct.

(4) Only Statement-II correct.

26) Functions of conducting part :

(1) Clears foreign particles

(2) Humidifies

(3) Brings the air to atmospheric temperature.

(4) Both (1) and (2)

27) Incomplete - Cartilaginous rings will be present in :

(1) Tracheae

(2) Tertiary Bronchi

(3) Initial bronchioles

(4) All of the above

28) Match the following –

(1)	Earthworm	(a)	Cutaneous respiration
(2)	Molluscs	(b)	Tracheal tubes
(3)	Insects	(c)	Moist cuticle

(4)	Frog	(d)	Branchial respiration

(1) 1-a, 2-c, 3-b, 4-d

(2) 1-c, 2-d, 3-a, 4-b

(3) 1-c, 2-d, 3-b, 4-a

(4) 1-a, 2-b, 3-d, 4-c

29) In which disorder respiratory surface is decreased due to smoking.

(1) Asthma

(2) Occupational respiratory Disorder

(3) Emphysema

(4) Bronchial Asthma

30) In lungs, gaseous exchange is done by :-

(1) With help of ATP

(2) By diffusion

(3) Fascilitated diffusion

(4) Both 1 & 2

31) Chemosensitive are is sensitive for :-

(1) O_2, CO_2

(2) CO_2, Temperature

(3) H^+, CO_2, O_2

(4) H^+, CO_2

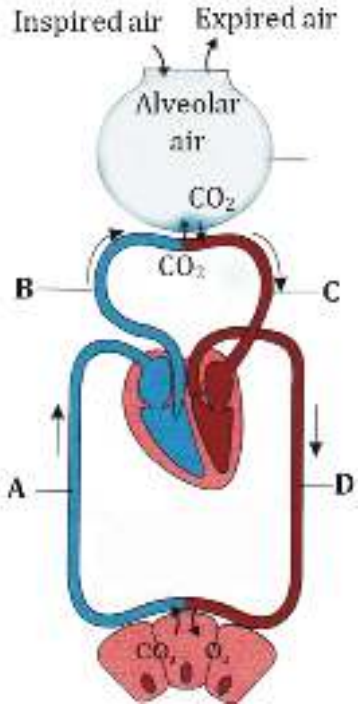
32) Respiratory rhythm centre present at :-

(1) Pons

(2) Medulla oblongata

(3) Cerebellum

(4) Cerebrum



33) Name the blood vessels A to D-

	A	B	C	D
(1)	Systemic vein	Pulmonary artery	Pulmonary vein	Systemic artery
(2)	Systemic artery	Systemic vein	Pulmonary artery	Systemic vein
(3)	Pulmonary artery	Systemic vein	Pulmonary vein	Systemic artery
(4)	Systemic vein	Pulmonary vein	Pulmonary artery	Systemic artery

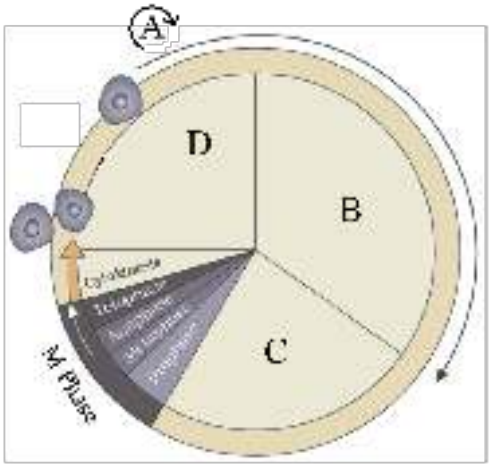
(1) 1

(2) 2

(3) 3

(4) 4

34) Given below is diagrammatic view of cell cycle indicating formation of two cell from one cell.



Mark the correct one for the events of this cell cycle :-

(1)	A	G_0	Cells is metabolically inactive

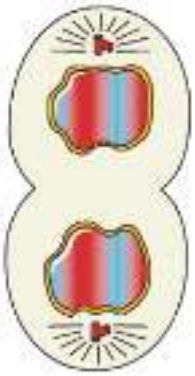
(2)	B	S	Amount of DNA per cell doubles but chromosome number remain same
(3)	C	G_2	Interval between mitosis and initiation of DNA replication where cell continuously grows
(4)	D	G_1	Shortest phase of cell cycle in which most of the organelles duplicate

(1) 1

(2) 2

(3) 3

(4) 4



35) Identify the above figure with their respective event :-

(1) Prophase ⇒ condensation of chromatin start

(2) Telophase ⇒ Centromere split and chromatids separate

(3) Metaphase ⇒ Initiation of the assembly of mitotic spindle

(4) Telophase ⇒ Nuclear envelope assembles around the chromosomes clusters.

36) Number of chromosome doubles in which stage of cell division :-

(1) Metaphase

(2) Telomere

(3) Anaphase

(4) Prophase

37) Synthesis of RNA & tubulin protein occur in :-

(1) Pre-mitotic gap phase

(2) Post-mitotic gap phase

(3) Synthesis phase

(4) M-phase

38) A. Centromere split and chromatids seprate
B. Chromatids move to opposite poles.
These statements are correct for which stage of cell division :-

(1) Interkinesis

(2) Interphase

(3) Telophase

(4) Anaphase

39) Meiosis ensures the production of _____ phase in life cycle of sexually reproducing organisms, where as fertilization restores _____ phase

(1) diploid, haploid
(2) haploid, triploid
(3) diploid, triploid
(4) haploid, diploid

40) At which stage of cell cycle, organelles like mitochondria and plastids get distributed between the two daughter cells ?

(1) Anaphase
(2) Metaphase
(3) Interphase
(4) Cytokinesis

41) X-shape structure (Chiasmata) is a characteristic feature observed during :

(1) Synaptonemal complex in zygotene stage
(2) Dissolution of the synaptonemal complex in diplotene stage
(3) Terminalisation of chiasmata in diakinesis stage
(4) Nucleolus disappears in diakinesis stage

42) Which of the following statement is correct :-

(1) In oocytes of some vertebrates, diplotene can last for months or years
(2) In oocytes of some vertebrates, diakinesis can last for months or years
(3) In oocytes of some vertebrates, pachytene can last for months or years
(4) In oocytes of some vertebrates, zygotene can last for months or years

43) Arrange the following events of meiosis in **correct** sequence :-
(a) Crossing over
(b) Synapsis
(c) Terminalisation of chiasmata
(d) Disappearance of nucleolus

(1) b → c → d → a
(2) d → a → b → c
(3) b → a → c → d
(4) a → b → c → d

44) Systematic different from taxonomy in considering

(1) Identification
(2) Nomenclature
(3) Phylogeny

(4) Classification

45) Which of the following pair is incorrect matched ?

(1) Mangifera indica - Species
(2) Homonidae - Family
(3) Diptera - Class
(4) Gymnospermae - Division

46) Modern taxonomic studies are based on

(1) Cell structures
(2) External and internal structure
(3) Ecological information of organisms
(4) All of these

47) **Assertion :** Potato and brinjal are two different species but both belong to the genus *Solanum*.
Reason : Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

48) Scientific name of Mango plant is *Mangifera indica* Linn. in the above name Linn. refers to :-

(1) Variety of Mango
(2) A taxonomist who proposed the present nomenclature in honour of linnaeus
(3) A scientist who for the first time described Mango plant
(4) A scientist who changed the name proposed by Linnaeus and proposed present name

49) Homonidae is equivalent to which category in taxonomic hierarchy of mango ?

(1) Sapindales
(2) Mangifera
(3) Anacardiaceae
(4) Dicotyledonae

50) Genera like Petunia and Mangifera is placed in

(1) Different family, order and class
(2) Different class but same divsion
(3) Same phylum and class
(4) Same Class but different order

51) ICBN stands for :-

(1) Indian Congress of Biological Name
(2) International Code of Botanical Nomenclature
(3) International Congress of Biological Name
(4) Indian Code of Botanical Nomenclature

52) In Binomial system when name of any plant is hand written then it should be :-

(1) Italicized
(2) Underlined cummulatively
(3) Underlined separately
(4) Closed in bracket

53) Which of the following name would be acceptable ?

(1) *Mangifera mangifera*
(2) *Solanum tuberosum* Linnaeus
(3) *Rattus rattus*
(4) All of the above

54) The classification system of Linnaeus was mainly based on the androecium structure therefore it was a/an :-

(1) Natural classification
(2) Numerical classification
(3) Artificial classification
(4) Phylogenetic classification

55) In which of the following system of classification each character (vegetative and sexual) is given equal importance ?
(a) Artificial system
(b) Numerical system
(c) Natural system
(d) Phylogenetic system

(1) a, d
(2) only b
(3) a, b and c
(4) a and b

56) The correct sequence of evolution in plants is :-

(1) Bryophyta → Gymnosperm → pteridophyta → Dicots → Monocots
(2) Bryophyta → Pteridophyta → Gymnosperm → Dicots → Monocots
(3) Pteridophyta → Gymnosperms → Dicots → Monocots → Bryophyta

(4) Bryophyta → Pteridophyta → Gymnosperm → Monocots → Dicots

57)

Which of the following is correctly matched with its particular taxonomic category :

(1) Mango – Sapindales is class
(2) housefly – Insecta is phylum
(3) Wheat – Triticum is genus
(4) Human – Carnivora is order

58) Which of the following classification is now easily carried out using computers is based on all observabale characteristics ?

(1) Natural classification
(2) Artificial classification
(3) Practical classification
(4) Numerical classification

59) Choose the correct descending sequence of taxonomic category:

(1) Class–Phylum–Order–Family–Genus–Species
(2) Phylum–Class–Order–Family–Genus–Species
(3) Species–Genus–Phylum–Class–Order–Family
(4) All of these

60) Match Column–I with Column–II for housefly classification and select the correct option using the codes given below:

Column I		**Column II**	
(a)	Family	(i)	Diptera
(b)	Order	(ii)	Arthropoda
(c)	Class	(iii)	Muscidae
(d)	Phylum	(iv)	Insecta

(1) a-iv, b-iii, c-ii, d-i
(2) a-iv, b-ii, c-i, d-iii
(3) a-iii, b-i, c-iv, d-ii
(4) a-iii, b-ii, c-iv, d-i

61) The group of organisms which are capable of interbreeding and producing fertile offsprings belongs to:

(1) Taxonomic species
(2) Biological species
(3) Specific epithet

(4) Subspecies

62) **Statement-I :** Members of a species have morphological similarities.
Statement-II : Members of same species are reproductively isolated.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and statement-II are incorrect
(3) Statement-I is incorrect and statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

63)

The number of species that are known and described range between :-

(1) 1.5 – 1.6 million
(2) 1.4 – 1.5 million
(3) 1.7 – 1.8 million
(4) 1.9 – 2.0 million

64) Who classified plants into trees, shrubs and herbs ?

(1) Aristotle
(2) Mendle
(3) linnaeus
(4) John Ray

65) *Chlorella, Chlamydomonas* and *Paramoecium, Amoeba* were earlier placed with plants and animals respectively but after Whittaker's 5 kingdom classification, they should be brought together in:-

(1) Monera
(2) Protista
(3) Plantae
(4) Animalia

66) The drawbacks or limitations of two-kingdom classification are

(1) No distinction between eukaryotes and prokaryotes
(2) No distinction between unicellular and multicellular organism
(3) Photosynthetic and non-photosynthetic organism are placed together
(4) All of the above

67) Which of the following criteria for 5 kingdom classification is not used by R.H. Whittaker-

(1) Mode of Nutrition
(2) Cell structure
(3) Reproduction

(4) Cell wall

68) The group of heterotrophic, eukaryotic, multicellular organisms lacking cell wall will be

(1) Monera
(2) Protista
(3) Plantae
(4) Animalia

69) **Statement-1 :-** The three-domain system divides the Kingdom Monera into three domains.
Statement-2 :- Five Kingdom system of Classification proposed by R.H. Whittaker.

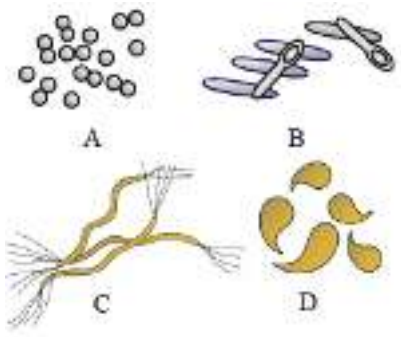
(1) Only statement-1 is correct
(2) Only statement-2 is correct
(3) Both statement-1 and statement-2 are correct
(4) Both statement-1 and statement-2 are incorrect

70) According to "Carl Woese" archaebacteria belong to domain–

(1) Monera
(2) Protista
(3) Plantae
(4) Archaea

71) Most extensive metabolic diversity is found in :-

(1) Members of Protista
(2) Members of Monera
(3) Members of fungi
(4) Members of Plantae

72) 

Select option showing correct identification for above figure from A to D :

	A	**B**	**C**	**D**
(1)	Bacillus	Coccus	Spirillum	Vibrio
(2)	Coccus	Bacillus	Spirillum	Vibrio
(3)	Coccus	Spirillum	Bacillus	Vibrio

(4)	Coccus	Vibrio	Bacillus	Spirillum

(1) 1
(2) 2
(3) 3
(4) 4

73) Genetic material without nuclear membrane is found in

(1) Eubacteria & Mycoplasma
(2) Cynobacteria & Eubacteria
(3) Archaebacteria & Blue green algae
(4) All of the above

74) Bacterial cell envelope includes-

(1) Glycocalyx only
(2) Cell wall only
(3) Glycocalyx and Nucleoid
(4) Glycocalyx, cell wall and cell membrane

75) How many are the correct statement about bacteria
(A) Bacteria are the sole member of kingdom monera.
(B) They are the most abundant microorganisms.
(C) They also live in extreme habitats.
(D) Bacteria are the member of kingdom protista.

(1) Three
(2) Two
(3) One
(4) Four

76) Which of the following is not a function of mesosome?

(1) Respiration
(2) Cell wall formation
(3) Protein synthesis
(4) DNA replication and its distribution in daughter cells

77) Which of the following is not a feature of the organisms of kingdom Monera?

(1) Presence of nucleoid
(2) Double stranded circular DNA
(3) Absence of membrane-bound organelles
(4) Cell wall made up of cellulose

78) **Assertion :** Membrane bound cell organelles are absent in prokaryotes.

Reason : Plasma membrane is absent in prokaryotes.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

79) Find out the correct match from the following table:-

	Coloumn I	**Coloumn II**
(i)	Fimbriae	Cell movement
(ii)	Capsule	Protein Synthesis
(iii)	Pili	Help in genetic recombination

(1) (i) and (ii)
(2) (ii) and (iii)
(3) only (iii)
(4) (i) and (iii)

80) A plasmid :-

(1) Is always attached with genophore
(2) Can replicate independently
(3) Can not replicate
(4) Is Main chromosome

81)

Match the following & choose the correct option :-

(A)	Family	**(i)**	Triticum aestivum
(B)	Kingdom	**(ii)**	Poales
(C)	Order	**(iii)**	Triticum
(D)	Species	**(iv)**	Plantae
(E)	Genus	**(v)**	Poaceae

(1) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv), (E)-(v)
(2) (A)-(v), (B)-(iv), (C)-(ii), (D)-(i), (E)-(iii)
(3) (A)-(iii), (B)-(ii), (C)-(i), (D)-(iv), (E)-(v)
(4) (A)-(iii), (B)-(v), (C)-(i), (D)-(ii), (E)-(iv)

82) What is A, B, C and D in given table ?

	Family	**Order**	**Class**	**Division /Phylam**

Potato	Solanaceae	**A**	Dicotyledonae	Angiospermae
Cat	Falidae	Carnivora	**B**	Chordata
House fly	**C**	Diptera	Insecta	Arthropoda
Tiger	Falidae	Carnivora	Mammalia	**D**

(1) A : Polymoniales; B : Mammalia;
C : Muscidae; D : Chordata

(2) A : Sapindales; B : Vertebrata;
C : Muscidae, D : Chordata

(3) A : Polymoniales; B : Mammalia;
C : Muscidae; D : Animalia

(4) A : Polymoniales; B : Primata;
C : Muscidae ; D : Chordata

83) Read the following statement A and B and choose the correct option :
Statement A : The number of common characteristics goes on increasing from species to kingdom
Statement B : Higher the category, greater is the difficulty of determining the relationship to other taxa of same level.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

84) Two kingdom classification system used for a long time was inadequate, so a need was felt for including besides gross morphology, other characteristics also. What were these characters which resulted in formation of five kingdom classification system ?
(a) Cell structure
(b) Mode of nutrition
(c) Reproduction
(d) Evolutionary relationship
Options :

(1) a & b only

(2) a, b & c only

(3) a, c & d only

(4) a, b, c & d

85) Which one of the following structure not found on the outer surface of bacterial cell.

(1) Capsule

(2) Mesosomes

(3) Pili

(4) Flagella

86) (A) Cell metabolically inactive.

(B) Chromosome condensation completed.
(C) Golgi body reappears
(D) DNA replication
(E) Number of chromosomes becomes double.
How many statements given above are **correct** for metaphase ?

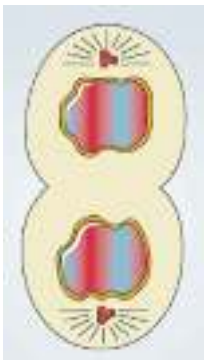
(1) 1

(2) 3

(3) 4

(4) 5

87) Identify the correct match from the column-I, II and III.

Column I		**Column II**		**Column III**	
A	Late prophase	a.		i.	ER, Golgi complex disappear
B	Anaphase	b.		ii.	Chromosome decondense and loose their individuality
C	Telophase	c.		iii.	Longest phase of meiosis-I
D	Prophase-I	d.		iv.	Splitting of centromere occur

(1) B-a-iii, A-c-iv, C-b-i, D-d-ii

(2) A-b-i, B-a-iv, C-d-ii, D-c-iii

(3) A-d-iii, B-b-iv, C-b-i, D-d-ii

(4) B-a-iii, A-c-iv, C-a-ii, D-d-i

88) Centrioles undergo duplication during ....(i).... of ....(ii)...., and begin to move towards opposite poles of the cell during ....(iii).... stage of ....(iv)....

	(i)	(ii)	(iii)	(iv)
(1)	S phase	Interphase	Prophase	Mitosis
(2)	S phase	Interphase	Anaphase	Mitosis
(3)	Prophase	Mitosis	Metaphase	Mitosis
(4)	Prophase	Mitosis	Anaphase	Mitosis

(1) 1

(2) 2

(3) 3

(4) 4

89) At which stage of meiosis, do these events occurs ?

	Bivalents more clearly visible	Formation of synaptonemal complex	Terminalisation of chiasmata
(1)	Leptotene	Zygotene	Pachytene
(2)	Zygotene	Pachytene	Diakinesis
(3)	Pachytene	Zygotene	Diakinesis
(4)	Zygotene	Diplotene	Diakinesis

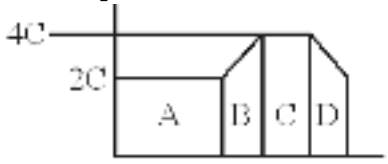
(1) 1

(2) 2

(3) 3

(4) 4

90) Identify the phases from the graph given below that shows the change in DNA content during various phases (A to D) of mitotic cell cycle.



	A	B	C	D
(1)	G_2	G_1	S	M
(2)	G_2	S	G_1	M
(3)	G_1	S	G_2	M
(4)	M	G_1	S	G_2

(1) 1

(2) 2

(3) 3

(4) 4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	1	1	1	1	2	2	3	4	2	2	3	3	1	3	2	3	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	3	1	2	3	1	3	4	1	4	4	3	3	1	3	2	4	1	4
Q.	41	42	43	44	45															
A.	1	1	3	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	1	1	4	3	2	2	1	3	3	1	3	2	3	4	3	3	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	1	3	4	3	1	2	4	3	3	1	3	1	2	3	3	2	1	3
Q.	86	87	88	89	90															
A.	2	2	4	1	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	1	2	1	4	2	1	4	1	3	2	2	2	1	2	4	1	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	3	1	1	4	4	3	3	2	4	2	1	2	4	3	1	4	4	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	3	3	3	4	1	3	3	4	2	3	3	3	4	2	3	4	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	3	1	2	4	4	4	2	4	2	2	4	4	1	3	4	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	1	2	4	2	1	2	1	3	3										

SOLUTIONS

PHYSICS

1) **Explanation Question:** Differentiation of different functions
Concept: This question is based on differentiation
Solution:

(I) $\frac{d(x\ell nx)}{dx} = \frac{xd\ell nx}{dx} + \ell nx\frac{dx}{dx}$

$= x\frac{1}{x} + \ell nx$

$= 1 + \square n\ x$

(II) $y = \frac{x}{2x+1} \Rightarrow \ell ny = nx - \ell n(2x+1)$

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x} - \frac{4}{2x+1} = \frac{1}{x(2x+1)}$

$\square \frac{dy}{dx} = \frac{1}{(2x+1)^2}$

(III) $y = 6x^2 \Rightarrow \frac{dy}{dx} = 12x$

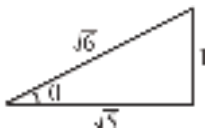
$\Rightarrow \frac{d^2y}{dx^2} = 12$

(IV) $y = \left(\frac{1}{2x+1}\right)$

$\frac{dy}{dx} = \frac{-2}{(2x+1)^2}$

Final Answer
The correct option is (1)

2) **Question Explanation:** To find the net displacement of body.
Concept: Distance and displacement.
Formula: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ & $r = \sqrt{x^2 + y^2 + z^2}$
Solution:
$\vec{A} = 6\hat{j}$
$\vec{B} = 8\hat{i}$
$\vec{C} = 10\hat{k}$
Net displacement $\vec{S} = \vec{A} + \vec{B} + \vec{C}$
$\vec{S} = 6\hat{j} + 8\hat{i} + 10\hat{k}$
or $|\vec{S}| = \sqrt{(8)^2 + (6)^2 + (10)^2}$
S = $10\sqrt{2}$m

3)  $\Rightarrow \cos\theta = \frac{\sqrt{5}}{\sqrt{6}}$

4) $\int_{-1}^{+1} t^{-3}.dt = \left[\frac{t^{-2}}{-2}\right]_{-1}^{+1}$

$= \frac{-1}{2} + \frac{1}{2} = 0$

5) $\cos 425° = \cos(360 + 65) = \cos 65°$

6) cosec (330°) = cosec(360° – 30°) cosec (330°) = – cosec30° = –2

7) $\frac{d(AB)}{dx} = \frac{AdB}{dx} + \frac{BdA}{dx} = x^2e^x + e^x2x$

$AB = \sqrt{(9-6)^2 + (12-7)^2 + (5-2)^2}$

$= \sqrt{(3)^2 + (5)^2 + (3)^2}$

$= \sqrt{9 + 25 + 9}$

8) $= \sqrt{43}$

9)

sin (45 + θ) cos (15 + θ) – cos (45 + θ) sin (15 + θ)
= sin [(45 + θ) – (15 + θ)]
= sin 30° = $\frac{1}{2}$

10)

a = sin 300°(4th quadrant)
b = cos 120° (2nd quadrant)

11) $v = \frac{4}{3}\pi r^3$
$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(1)^2 3$
$= 12\pi \frac{m^3}{sec}$

12) $y = 2\sin^2\theta + \tan\theta$
$\frac{dy}{d\theta} = \frac{d}{d\theta}[2\sin^2\theta + \tan\theta]$

$= \frac{d}{d\theta}[2\sin^2\theta] + \frac{d}{d\theta}[\tan\theta]$

$= 2\frac{d}{d\theta}[\sin^2\theta] + \sec^2\theta$

$= 2\,(2\,(\sin\theta) \times \frac{d}{d\theta}(\sin\theta) + \sec^2\theta$

$= 2[2\sin\theta\cos\theta] + \sec^2\theta$

$= 2[\sin 2\theta] + \sec^2\theta$

13) $y = \frac{1}{x+1}$

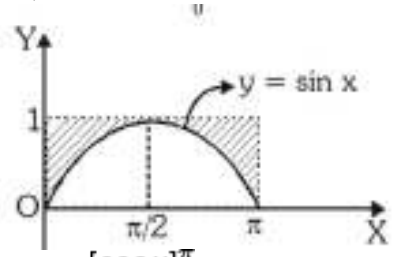
$y = (x+1)^{-1}$

$\frac{dy}{dx} = -1\,(x+1)^{-2}$

$\frac{dy}{dx} = -\frac{1}{(x+1)^2}$

14) $\frac{x^3}{3} - \sin x + \log_e x + c$

15) Area = $1 \times \pi - \int_0^{\pi} \sin x\, dx$



$= \pi + [\cos x]_0^{\pi}$

$= \pi + [\cos\pi - \cos 0]$

$= \pi + [-1 - 1]$

$= \pi - 2$

16) 12

17) $y = \frac{2}{3}x - \frac{5}{3}$

Slope = +ve

y – intersect = –ve

18) **Explaining :** to find integration

Concept : This question is based on

$\int_0^{25} y.dx =$ area under the curve

Solution :

$$\int_{x_1}^{x_2} y.dx =$$ Area under curve

$$\int_{0}^{25} y\ dx = \frac{1}{2} \times [10 + 25] \times 20$$

$$\int_{0}^{25} y.dx = 350$$

Final Answer : (3)

19) **Explain Question:**
The question asks to match the columns with correct Affirmations.

Concept:
Mixed Differentiation + integration.

Solution:

(P) $y = \frac{d}{dx}(x^2 + 3x - 4)$

$y = 2x + 3$

At $x = 0 \Rightarrow y = 3$

(Q) $y = \frac{d}{dx}[3\sin x - 4\cos x]$

$y = 3\cos x + 4\sin x$

$at\ x = \frac{\pi}{2} \Rightarrow y = 3(0) + 4 = +4$

(R) $y = \int_{0}^{1} (x^2 - 2x + 1)dx$

$y = \left[\frac{x^3}{3} - x^2 + x\right]_0^1$

$y = \frac{1}{3} - 1 + 1 = \frac{1}{3}$

(S) $y = \int_{-\infty}^{0} e^x dx$

$y = [e^x]_{-\infty}^{0} = e^0 - e^{-\infty} = 1 - 0$

$y = 1$

Final Answer: option(3)

20)

Explain Question : In this question we have to choose the correct graph for the equation $y = 3x^2 - 5$

Concept : This question is based on coordinate geometry and some standard graph.

Solution : For a quadratic equation of the form
$y = ax^2 + 6x + c$
The graph is parabola with mouth opening towards ± y -axis [depends on 'a']
Given here : $y = -3x^2 - 5$
here a = –3, therefore the graph is parabola with mouth opening toward negative y-axis at x = 0, y = –5 → starting point.
Therefore the correct graph is represented in option (1).

Final answer : Option (1)

21)

Explain Question : We are asked to find the minimum value of the function $x = 3\sin\theta - 4\cos\theta + 5$.
Concept : This question is based on Minimum value of a trigonometric function.
Solution : The minimum value of an expression of the form
$a\sin\theta + b\cos\theta$ can be given by
$R_{minimum} = -\sqrt{a^2 + b^2}$
Given here
$x = 3\sin\theta - 4\cos\theta + 5$
$\therefore X_{minimum} = R_{minimum} + 5 = -\sqrt{a^2 + b^2} + 5$
here, a = 3, b = – 4
$\therefore X_{minimum} = -\sqrt{(3)^2 + (-4)^2} + 5$
$= -\sqrt{25} + 5$
= – 5 + 5 = 0
Final answer : Option (1)

22) $S_n = \frac{n}{2}[1^{st}\text{term} + n^{th}\text{term}]$ $a + (n-1)d = 33$
$S_n = \frac{17}{2}[1 + 33]$ $n = 17$ $S_n = 289$

23) A + B = 17
A – B = 7
A = 12 B = 5
$R = \sqrt{A^2 + B^2} = 13$

24)

Projection of $\vec{A}$ on $\vec{B}$ = $\left(\frac{\vec{A} \cdot \vec{B}}{B^2}\right)\vec{B}$

here $\vec{A} \cdot \vec{B} = 8$

$\left|\vec{B}\right| = \sqrt{3}$

So projection of $\vec{A}$ on $\vec{B} = \frac{3}{8}\left(\hat{i} + \hat{j} + \hat{k}\right)$

25)

Explain question:
The question asks to find the magnitude of resultant force.

Concept:
Vector addition.

Solution:
Angle between both the force is = 120°
(using parallel shifting of vectors)

$R = 2F\cos\frac{\theta}{2}$

$R = 2(25)\cos\frac{120}{2} = 50\cos 60°$

$R = 50\left(\frac{1}{2}\right) = 25\,N$

Final answer: Option (2)

26) $\left|\hat{A} + \hat{B}\right| = \sqrt{3}$ (A=1, B=1)

$2 + 2\cos\theta = 3 \Rightarrow \theta = 60°$

$\Rightarrow (2\hat{A} + 3\hat{B}) \cdot (\hat{A} + 3\hat{B})$

$= (2A^2 + 3\hat{A} \cdot \hat{B} + 6\hat{A} \cdot \hat{B} + 9B^2)$

$= 2A^2 + 9AB\cos 60 + 9B^2$

$= 2 + \frac{9}{2} + 9 \Rightarrow \frac{31}{2}$

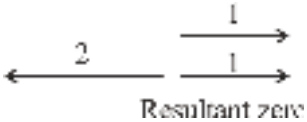
27) Resultant of $\vec{A}$ and $\vec{B}$

$\vec{R} = \vec{A} + \vec{B} = 4\hat{i} + 3\hat{j} + 6\hat{k} - \hat{i} + 3\hat{j} - 8\hat{k}$

$\hat{R} = \frac{\vec{R}}{\left|\vec{R}\right|} = \frac{3\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{3^2 + 6^2 + (-2)^2}}$

$= \frac{3\hat{i} + 6\hat{j} - 2\hat{k}}{7}$

28) $A - B \le R \le A + B$



29) $\cos\theta = \frac{\vec{A}.\vec{B}}{AB}$

$= \frac{42+24-12}{(9)(9)} = \frac{54}{81} = \frac{2}{3}$

$\sin\theta = \sqrt{1-\frac{4}{9}} = \frac{\sqrt{5}}{3}$

$\theta = \sin^{-1}\frac{\sqrt{5}}{3}$

30) Here $\vec{A}\cdot\vec{B} = 0$

$3 + 2\alpha + 6\alpha = 0$

$8\alpha = -3$

31) $\vec{A}\times\vec{B} = \vec{R}$

The direction of $\vec{R}$ would be perpendicular to plane containing $\vec{A}$ and $\vec{B}$

32) $ab\sin\theta = \frac{1}{\sqrt{3}}ab\cos\theta$

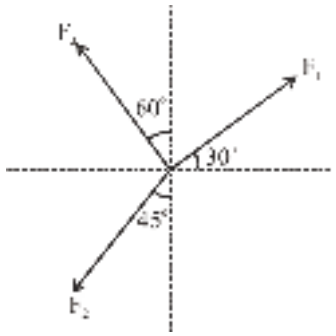
$\tan\theta = \frac{1}{\sqrt{3}}$

$\theta = 30°$

$\left|\vec{a}+\vec{b}\right| = \sqrt{a^2+b^2+2ab\ \cos 30^o}$

$\left|\vec{a}+\vec{b}\right| = \sqrt{a^2+b^2+\sqrt{3}ab}$

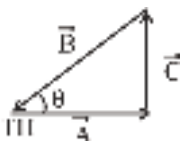
33)



$F_2\cos45^o = F_1\sin30º + F_3\cos60^o$

$\sqrt{2}F_2 = F_1 + F_3$

34)



$\sin\theta = \frac{C}{B} = \frac{1}{2} \Rightarrow \theta = 30°$

Angle between $= 180° - \theta = 180° - 30° = 150°$

35) $3\vec{a} + \vec{b} = 4\hat{i} + 5\hat{j} + 7\hat{k}$

$|3\vec{a} + \vec{b}| = \sqrt{(16 + 25 + 49)} = 3\sqrt{10}$ unit.

36) (1)-S (2)-P (3)-Q (4)-R

37) Displacement = 2r = 20 cm

38) $\text{distance} = \pi r$

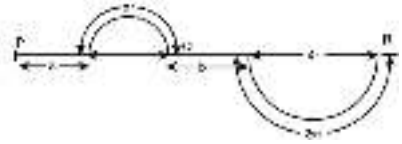
displacement = 2r

$$\frac{\text{distance}}{\text{displacement}} = \frac{\pi}{2}$$

39)

25 s

40) a = 7m, b = 8m, $r = \frac{11}{\pi}\left[\pi = \frac{22}{7}\right]$



Distance travelled by the car from P to R

$= a + \pi r + b + 2\pi r$

$= a + b + 3\pi r$

$= 7 + 8 + 3\pi \times \frac{11}{\pi} = 48\text{m}$

41) $2R\sin\theta/2$

$|\overrightarrow{AB}| = 2R\sin(\theta/2)$

42) $\vec{d} = 20\hat{j} + 30\hat{i} + 40\sqrt{2}\left(\frac{-\hat{i} - \hat{j}}{\sqrt{2}}\right)$

$\vec{d} = -10\hat{i} - 20\hat{j}$

$\tan\theta = \frac{20}{10} \Rightarrow \theta = \tan^{-1}(2)$

$10\sqrt{5}\,\text{m}$, $\tan^{-1}(2)$ S of W

43) $|\vec{d}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

$\hat{i} + \hat{j}$

44) distance = $\frac{2\pi R}{4} = \frac{\pi R}{2}$

displacement = $\sqrt{R^2 + R^2} = \sqrt{2}R$
$5\sqrt{2}\,m, 2.5\,\pi\,m$

45) 1rev → 40 sec → 2πR
T = 2 min + 20 sec
= 3 × 40 + 20 sec
$= 3 \text{ rev} + \frac{1}{2} \text{ rev}$
$= 0 + \boxed{2R}$

CHEMISTRY

46) n = 4, means electron is in 4th shell and = 2, means subshell is d. Therefore, the orbital is in 4d-subshell.

47)

In hydrogen (one electron), energy depends only on **principal quantum number (n).**
Final Answer : option (1)

48) l cannot be greater than n

49)

's' orbital is spherical in shape.

50) (a) n = 2, l = 1 corresponds to the 2p orbital.

(b) n = 3, l = 2 corresponds to the 3d orbital.

(c) n = 3, l = 0 corresponds to the 3s orbital.

(d) n = 2, l = 0 corresponds to the 2s orbital.
This gives the sequence of matches: (a) (iii), (b) - (iv), (c) (ii), (d) - (i).
Final answer : 4

51) Conceptual

52)

Size → depends on n
Shape → depends on
Orientation of orbital → depends on m
Spin of electron → depends on s

53) Orbital angular momentum = $\sqrt{l(l+1)}\frac{h}{2\pi}$
For the 2s orbital, the quantum number □ is 0:
Orbital angular momentum = $\sqrt{0(0+1)}\frac{h}{2\pi} = 0$
For the 2p orbital, the quantum number □ is 1:
Orbital angular momentum = $\sqrt{1(1+1)}\frac{h}{2\pi} = \sqrt{2}\frac{h}{2\pi}$
Final Answer : (2)

54) Because only in first configuration two electron have all four quantum number identical.

55) (d) < (b) < (c) < (a)

56) **Concept** : For a given principal quantum number (n), the total number of orbitals is given by the formula n^2.
Solution : Number of orbitals in a shell $n^2 = (3)^2 = 9$
Final Answer : Option (3)

57)
$n + \ell = 7$
$7 + 0 = 7s$
$6 + 1 = 6p$
$5 + 2 = 5d$
$4 + 3 = 4f$
] 4subshell

58) For a given value of n,
□ = 0 to (n – 1)
and for a given value of □,
m = –□ to + □
$s = +\frac{1}{2} or -\frac{1}{2}$
Hence, set (ii) (n = 3 and □ = 3), set (iii) (s = +1) and set (iv) (s = 0) are not possible.

59)

Solution/Explanation:
We need to find expression of maximum no. of e^- in a subshell.
No. of e^- in subshell = 2 (2□ +1)
= 4□ + 2 .
Hence the correct answer is option (2)

60)

In a hydrogen atom, the energy levels of different subshells within the same principal quantum number are degenerate, meaning they have the same energy. This is because the energy levels in a hydrogen atom depend primarily on the principal quantum number (n) and not on the azimuthal quantum number (l).
Since all three subshells (3s, 3p, and 3d) belong to the same principal quantum number (n=3), they have the same energy level. Therefore, $\mathbf{E_1 = E_2 = E_3}$.

61) 
nf–orbital has $2e^-$ with Opposite spin.

62) **Explanation:-** n, ℓ and m value of an electron in $3p_y$ orbital are. **concept :-** n =1, 2, 3 ,..
ℓ = 0 to n-1, ℓ = 0; s - sub shell
ℓ = 1; p-sub shell
ℓ = 2; d- sub shell
ℓ = 3; f- sub shell
m = -ℓ to + ℓ, S = $+\frac{1}{2}$ or $-\frac{1}{2}$
for $3p_y$ orbital.
n = 3, ℓ = 1, m = -1 to +1 S = $+\frac{1}{2}$ or $-\frac{1}{2}$
Answer :- so option (1) and option (2) correct.

63)

Nodal plane (angular node) = ℓ
ℓ = 2 (for d)

64) $n = 3,\ \ell = 1,\ m = -1$
Represent one orbital, hence can accomodate only two $e^{\ominus}$

65)

According to Aufbau principle the lower energy 2s orbital must be completely filled before e^- enters the 2p orbital.

66) Solution-
Energy of H-like species depends on Principal quantum number (n) only.

67) Conceptual

68)

Electronic Configuration of $_{24}Cr$ –
$1s^2\ \ 2s^2\ \ 2p^6\ \ 3s^2\ \ 3p^6\ \ 4s^1\ \ 3d^5$
Total number of electrons with (n = 3) = 3 + 6 + 5 = 13

Final Answer : (1)

69)

1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < ---(for single electronic species)

70)

Heisenberg Uncertainity principle is not applicable for big particles and stationary particles.
Answer option (4)

71)

Conceptual

72)

Spin angular momentum = $S = \sqrt{s(s+1)} \times \left(\frac{h}{2\pi}\right)$
Final answer : option (1)

73) **Question Explanation :**
We need to find which ion has μ = 2.83 BM.
Concept :
Magnetic moment (μ) = $\sqrt{n(n+2)}$. B. M. where n = no. of unpaired electrons.
Solution :
μ = 2.83 BM → n(n+2) = $(2.83)^2$ = 8 ≈ n=2.
So ion must have 2 unpaired electrons.
Check ions:

A. Ti^{3+} (Z=22 → $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$): 1 unpaired → μ ≈ 1.73 BM.
B. Ni^{2+} (Z=28 → [Ar]$3d^8$): 2 unpaired → μ ≈ 2.83 BM
C. Cr^{3+} (Z=24 → [Ar]$3d^3$): 3 unpaired → μ ≈ 3.87 BM.
D. Mn^{2+} (Z=25 → [Ar]$3d^5$): 5 unpaired → μ ≈ 5.92 BM.

Final Answer : Option (2)

74) All three relations are correct mathematical representations of Heisenberg's Uncertainty Principle.
The correct relations are all of the above.
Final Answer : All of the above

75)

Explanation Question :
We are comparing orbital energies based on the (n + ℓ) rule.
Concept :
This question is based on energy order is decided by (n + ℓ).

If (n + ℓ) is the same, then the orbital with lower n has lower energy.
Solution :
For 3d: n + ℓ = 3 + 2 = 5.
For 4p: n + ℓ = 4 + 1 = 5.
Since both are equal, the orbital with smaller n (3d) has lower energy than 4p.
So, Statement I is incorrect (3d < 4p).
Statement II is correct (rule explanation).
Final Answer : 3

76) $\Delta x \cdot \Delta p = \frac{h}{4\pi}$

$\Delta x = \frac{h}{4\pi \Delta p} = \frac{0.527 \times 10^{-34}}{10^{-5}}$

77) no. of waves = orbit no.

78)

Element and compound are pure substance.

79)

1 amu × N_A = 1g

80) $\frac{W(g)}{Mw} = \frac{V(L)}{22.4}$

$\frac{240}{64} = \frac{V}{22.4}$

V = 84 L

81)

(1) No of molecules of CO_2 = N_A × mole = $N_A \times \frac{44}{44} = N_A$

(2) No of molecules of O_2 = N_A × mole = $N_A \times \frac{48}{32}$ = 1.5 N_A

(3) No of molecules of H_2 = N_A × mole = $N_A \times \frac{8}{2}$ = 4 N_A

(4) No of molecules of SO_2 = N_A × mole = $N_A \times \frac{64}{64} = N_A$

82)

1 pico = 10^{-12}
Final Answer : option (3)

83)

S. I. unit of electric current = Ampere

84)

1 mol = 40 g
2.5 mol = 100 g

85) 1 amu = 1.67×10^{-24} g

86) $\frac{W(g)}{Mw} = \frac{V}{22.4}$

$\frac{1.25}{Mw} = \frac{1}{22.4}$

Mw = 28 g/mol

87)

Question Explanation : We have to convert a given pressure in mm of Hg into atmospheric pressure unit.

Concept : This question is based on Pressure unit conversion.

Solution :

 760 mm Hg = 1 atm

1 mm Hg = $\frac{1}{760}$ atm

3800 mm Hg = $\frac{1}{760} \times 3800 = 5$ atm.

3800 mm Hg = 5 atm.

Final Answer: option (2)

88)

Compounds have different property from its constituent elements from which it has been formed.
SI system has sever base units → second, metre, kilogram, ampere, kelvin, mole and candela.
→ SI unit of volume is cubic metre (m^3).

89)

Let mass of all gases = x

$n_{O_2} : n_{H_2} : n_{CH_4}$

$\left\{\frac{x}{32} : \frac{x}{2} : \frac{x}{16}\right\} \times 32$

1 : 16 : 2

90)

Mass of 1 H_2O molecule = 18 amu
$= 18 \times 1.67 \times 10^{-24}$ g
$= 30 \times 10^{-24}$ g
$= 3 \times 10^{-23}$ g

BIOLOGY

91)

Question Explanation: The question asks to evaluate the assertion and reason about echinodermata (spiny-skinned animals like starfish).

Concept: The question is based on Phylum Echinodermata

Solution: The correct answer is option 1

A. Assertion: Spiny skinned animals like starfish are called echinodermata → True. The name "echinodermata" literally means "spiny-skinned."

B. Reason: These animals have an endoskeleton of calcareous ossicles → True. Echinoderms possess an internal skeleton made of calcareous ossicles.

C. The Reason correctly explains the Assertion, because the spiny appearance is due to the calcareous ossicles forming the endoskeleton.

Final Answer: Option 1

92)

Question Explanation : Which statement about the occurrence of notochord in different animals is incorrect?

Concept: The question is based on chordata

Solution: The correct answer is option 3.

A. In humans, the notochord is **present during the embryonic stage** and later replaced by the vertebral column; So it is **not absent from the very beginning**.
B. In ascidians, the notochord is present only in the larval tail; in frogs it is replaced by the vertebral column in adults; and in amphioxus it persists throughout life.

Final Answer: Option 3

93)

Question Explanation: Evaluate two statements on mammalian features.

Concept: Mammalian characteristics and limbs.

Solution: Statement-I is correct; the presence of mammary glands is a defining and unique feature of the class Mammalia.
Statement-II is also correct; mammalian limbs are adapted for a wide variety of functions like walking, running, climbing, burrowing, swimming, or flying (e.g., bats).
Since both statements are correct, Option 1 is the answer.

Final Answer: Statement-I and II both are correct.

94)

Question Explanation : In which group of fishes are both scales and paired fins absent?

Concept: The question is based on Cyclostomes

Solution: The correct answer is option 2.

A. **Cyclostomes** (jawless fishes like *Petromyzon* and *Myxine*) lack **scales** and also **do not possess paired fins**.
B. Chondrichthyes and Osteichthyes have paired fins, and Osteichthyes also have scales.

Final Answer: Option 2

95) Just as the water vascular system of echinoderms serves multiple functions including respiration and help in excretion, the feather-like gills in molluscs also show multiple functions of respiration and excretion.

96) **Solution :** *Limulus*.commonly known as king crab belongs to phylum arthropoda. It is a living fossil and large sized marine arthropod.
Final Answer : (4)

97) **Solution :** Lampreys and hag fishes (*Maxine*) are unusual, jawless fish that comprise the order cyclostomata, so named because of the circular shape of the mouth. The brains of lampreys and hagfishes differ a lot, but they also show a large number of similarities, as do all craniate brains.
Final Answer : (2)

98) **Solution :** Aquatic annelids like *Nereis* possess lateral appendages (called parapodia) which help in swimming. Radula is the rasping organ for feeding which is present in the mouth of molluscs.
Final Answer : (1)

99) **Solution :**

	Column-I (Term/Feature)		Column-II (Examples)
A.	Gregarious pest	(i)	Hirudinaria
B.	Vector	(ii)	Plannaria
C.	Oviparous with indirect development	(iii)	Sepia
D.	Metameres	(iv)	*Aedes*
E.	High regeneration capacity	(v)	*Locust*

Final Answer : (4)

100) **Solution :** The given figures (A, B and C) are respectively hippocampus, catla (both are bony fishes) and salamander (amphibian). Hippocampus have a skeleton made up of bony plates, they use gills to breath and have an inflatable bladder to help regulate their buoyancy in the water. Catla is a member of ostoichthyes and possesses air bladder to regulate buoyancy.
Final Answer : (1)

101) **Solution :** Penguin, kiwi and ostrich all belong to class Aves (.i e. birds) under phylum chordata and they do not give birth to their young ones. They are oviparous while kangaroo, hedgehog, dolphin and loris, all belong to class mammalia and are viviparous.
Final Answer : (3)

102) **Solution : Chondrichthyes →** Chordata → Vertebrata → Cartilaginous skeleton → Placoid scales → No operculum
Final Answer : (2)

103)

Explain Question : Asking which phylum shows bilateral symmetry in larva and radial symmetry in adults.

Concept : Based on phylum Echinodermata

Explanation : The correct option is 2 : Echinodermata exhibit this symmetry change during metamorphosis.Larvae are bilaterally symmetrical and adults have pentamerous radial symmetry.

Final answer : 2 – Echinodermata

104) NCERT-XI, Pg # 55

105) NCERT XI, Page # 54 IInd para

106) NCERT XI, Page # 57 Ist & IInd para

107) **Question Explanation:** In amphibian, cloaca is common chamber for

Concept: Amphibia
Solution: The cloaca is a common chamber in amphibians for the alimentary canal, urinary tract, and reproductive tract.
Final Answer: 4

108) NCERT XI Pg.#52

109)

Question Explanation : : This question asks about the correct match between class, character and example.
Concept : This qeustion is based on Classes of Chordates.

Aves – Forelimbs are modified into wings → True feature, but in the given example (macropus, not a bird) it doesn't match correctly.
Reptilia – Tympanum represents ear → Correct, reptiles like lizards use tympanum as a hearing organ.
Mammalia – Air sacs connected to lungs → Wrong, this is a feature of Aves, not mammals.
Amphibia – Homeothermic → Wrong, amphibians are poikilothermic (cold-blooded).
Final answer : 2

110) NCERT XI Pg.# 271

111) NCERT, Pg No- 187

112) NCERT, Pg No- 186-187

113) NCERT, Pg - 189

114) **Concept :**
Solution : Contraction → volume → pressure → air in.
Final Answer : (1)

115) NCERT Pg. # 17.2

116) NCERT Pg. # 17.1.1

117) NCERT Pg. # 17.1.1

118) NCERT Pg. # 268 (17.1)

119) NCERT Pg. # 275 (17.6)

120) NCERT Pg. # 272

121) NCERT Pg. # 275

122) NCERT Pg. # 275

123)

Explain Question : Identify blood vessels A, B, C, D in diagram.

Concept : This question is based on Exchange of gases.

Solution :
A (from body → right atrium): systemic vein (deoxygenated blood).
B (from right ventricle → lungs): pulmonary artery (deoxygenated blood to lungs).
C (from lungs → left atrium): pulmonary vein (oxygenated blood).
D (from left ventricle → body): systemic artery (oxygenated blood to tissues).

Thus, the correct sequence is:
A = systemic vein, B = pulmonary artery, C = pulmonary vein, D = systemic artery.

Final Answer : option (1) A: Systemic vein, B: Pulmonary artery, C: Pulmonary vein, D: Systemic artery

124) **Question Explanation :**
Decide which given statements correctly match the cell-cycle phases A-D with their described
Concept :
Phases of cell cycle.
Solution :
(1) A → G_0 = metabolically inactive → False.
(2) B → S = DNA amount doubles, chromosome number same → True.
(3) C → labeled G_2 but described as "interval between mitosis and initiation of DNA replication" (that is actually G_1) → True.
(4) D → G_1 dose have organelle duplication but is not the shortest phase (statement claims it is shortest) → False.
Final Answer :
Option (2)

125)

Question Explanation:Identifying a stage of cell division.

Concept: Mitosis

Solution: The figure shows the final stage of mitosis. The chromosomes have reached the opposite poles of the cell, and new nuclear envelopes are forming around the two sets of chromosomes. This indicates the cell is in Telophase.

Final Answer:Option 4: Telophase -> Nuclear envelope assembles around the chromosomes clusters.

126) **Question Explanation**
The question is asking in which stage of cell division the number of chromosomes effectively doubles.
Concept
anaphase.
Solution
During anaphase, the sister chromatids of each chromosome separate and move to opposite poles. Each chromatid is now considered an individual chromosome, so the effective chromosome number doubles in anaphase.
Final answer: Option 3

127) **Explanation**

A. RNA Synthesis: RNA is synthesized throughout interphase (G1, S, and G2 phases) as it's needed for various cellular processes like protein synthesis, including the proteins needed for cell division.

Tubulin Protein Synthesis: Tubulin is the protein that forms microtubules, which are crucial for the mitotic spindle during cell division (M phase). Tubulin synthesis mainly occurs during the G2 phase (pre-mitotic gap phase), as the cell prepares for mitosis.

Correct answer:1. Pre-mitotic gap phase (G1 phase)

128)

Question explanation
The question is asking which stage of cell division involves splitting of centromeres and movement of chromatids toward opposite poles.

Concept
Mitosis – anaphase stage.

Solution
During anaphase, the centromeres split and sister chromatids separate and move to opposite poles of the cell. Interphase and interkinesis are non-dividing phases, while telophase involves reformation of nuclear membranes.

Final answer: Option 4

129) NCERT (XI) Pg. # 167

130) NCERT Pg.#124

131) NCERT-XI Pg.#168

132)

Question explanation
The question is asking which meiotic stage remains arrested for months or years in the oocytes of some vertebrates.

Concept
Meiotic I

Solution
In many vertebrates, including humans, oocytes remain arrested for a long period in the diplotene stage of prophase I. This stage is also known as the dictyotene stage.
Final answer: Option 1

133)

Correct answer: Option 3 - b → a → c → d
Correct Sequence of Events in Meiosis:
1. (b) Synapsis → Occurs in Zygotene

A. Homologous chromosomes start pairing (synapsis) and form the synaptonemal complex.

2. (a) Crossing over → Occurs in Pachytene

A. Exchange of genetic material takes place between homologous chromatids.
B. G_1 phase (Option 1) → Cell grows and prepares for DNA replication but does not synthesize spindle proteins.
C. S-phase (Option 2) → DNA replication occurs, but spindle fibre proteins are not produced.
D. G_2 phase (Option 3) → Spindle fibre proteins (like tubulin) are synthesized, preparing for mitosis.
E. Early Prophase (Option 4) → Spindle fibres start assembling, but their synthesis happens earlier in G_2 phase.

Thus, spindle fibre proteins are synthesized in the G_2 phase.

134) NCERT Pg.# 7,8

135)

NCERT Pg.# 11

136)

Question explanation: Question is asking about modern taxonomic studies based on which factor.

Concept: Taxonomy -definition

Solution: The correct answer is option 4.

A. External and internal structure, along with the structure of cell, development process and ecological information of organisms are essential and form the basis of modern taxonomic studies.

Final answer: Option 4

137)

Explain Question: The question evaluates an Assertion–Reason pair related to genus and species.

Concept: Genus and species

Solution: Potato (Solanum tuberosum) and brinjal (Solanum melongena) are different species but belong to the same genus (Solanum) → Assertion is true.

A genus is defined as a group of closely related species sharing more common characters with each other than with species of other genera → Reason is true. This definition directly explains why potato and brinjal are placed in the same genus.

Final answer:1

138)

Module

139)

Explain Question: The question asks which taxonomic category is equivalent to Hominidae.

Concept: Family

Solution: Hominidae represents a family, among the given options Anacardiaceae (family of mango) is equivalent to Hominidae.

Final answer: 3

140) NCERT Pg.# 10

141)

NCERT Pg. # 6

142)

NCERT (XI) Pg. # 4

143)

NCERT-XI, Pg. # 4

144) NCERT Page # 29

145)

Module-1, Pg. # 6

146)

Module-1, Pg. # 6

147)

NCERT XI, Pg # 11 ; Table : 1.1

148)

Module-1, Pg. # 7

149)

NCERT-XI, Pg. # 08

150)

NCERT-XI, Pg. # 8

151)

NCERT-XI, Pg. # 6

152)

NCERT-XI, Pg. # 6

153)

NCERT XI Pg. # 6

154) **Explain Question:** Question asks about who classified plants into trees, shrubs, and herbs.
Concept: This question is based on History of taxonomy
Solution: Morphological classification into trees, shrubs, and herbs →Aristotle

Final answer: 1

155)

Solution:

The correct answer is:2. Protista
According to Whittaker's 5 kingdom classification, organisms like Chlorella (alga), Chlamydomonas (alga), Paramoecium, and Amoeba (protozoans) are all unicellular eukaryotes and should be classified under Protista, which is a separate kingdom for such organisms.

156)

Question Explanation: Question asks about the major drawbacks of the old two-kingdom classification (Plantae and Animalia).

Concept: Two-kingdom classification

Solution: Two-kingdom system does not separate:
Prokaryotes vs. eukaryotes
Unicellular vs. Multicellular
Photosynthetic and non-photosynthetic organisms (e.g., Euglena)

Final answer: Option 4 (All of the above)

157)

Question Explanation: Question asks which criterion was NOT used by R.H. Whittaker while proposing the five-kingdom classification.

Concept: Five-kingdom classification

Solution: Whittaker used the following criteria: Cell structure , Mode of nutrition , Reproduction , Phylogenetic relationships But he did NOT use cell wall composition as a criterion.

Final answer: 4 (Cell wall)

158)

NCERT-XI, Pg. # 11

159)

Question Explanation: Question asks whether each statement about the three-domain system and Whittaker’s five-kingdom system is correct.

Concept: Three-domain system

Solution:Statement-1 :- The three-domain system divides the Kingdom Monera into three domains-false because Monera splits into two domains: Archaea and Bacteria. Statement. 2. Five-kingdom system was proposed by R.H. Whittaker (1969)-true.

Final answer: 2 (Only statement-2 is correct)

160)

NCERT-XI, Pg. # 11

161)

NCERT XI, Page No. # 19

162) **Explain Question :**The question asks to identify bacterial shapes A–D from the drawings and choose the matching option.
Concept : This question is based on Bacteria
Explanation :A shows spherical clusters → Coccus.B shows rod-shaped cells → Bacillus.C shows elongated curved cells with flagella → Spirillum.D shows comma-shaped cells → Vibrio.
Final answer : 2

163)

NCERT-XI, Pg. # 11

164)

Module-1, Pg. # 21

165) NCERT, Page No. 18

166)

Module-1, Pg. # 36

167)

NCERT-XI, Pg. # 13

168)

Question Explanation: Question asks about the truth of an assertion–reason pair related to prokaryotic cell structure.

Concept: Structure of bacteria

Solution:(A)Prokaryotes **lack membrane-bound organelles** (true).

(R)Prokaryotes **do have a plasma membrane**; it is essential for life (so the reason is false).

Final answer: Assertion is true but Reason is false.

169) NCERT XI Pg. # 129

170)

Module-1, Pg. # 23

171)

NCERT-XI, Pg. # 8

172) NCERT-XI, Pg.# 6, 7, 8

173)

NCERT-XI, Pg. # 8

174)

NCERT-XI, Pg. # 11

175) **Explain Question:** Question asks about which structure is not found on the outer surface of a bacterial cell.
Concept: This question is based on Structure of bacterial cell
Solution: Mesosomes: Infoldings of plasma membrane inside the cell, not on outer surface
Final answer: 2

176)

NCERT-XI, Pg. # 123

177) **Explanation**

A (Late prophase): (b) ER, Golgi complex disappear - During late prophase of meiosis, the endoplasmic reticulum (ER) and Golgi complex break down.

B (Anaphase): (a) Splitting of centromeres occurs - During anaphase centromeres split, allowing homologous chromosomes to separate.

C (Telophase): (d) Chromosomes decondense and lose their individuality - In telophase I, the chromosomes begin to decondense, and their distinct structure is lost.D (Prophase): (c)

Longest phase of meiosis I - Prophase I is the longest phase of meiosis and involves complex events like synapsis and crossing over.

Correct answer: 2. A-b-i, B-a-iv, C-d-ii, D-c-iii

178) NCERT XI Pg. # 164

179)

NCERT-XI, Pg. # 121

180) NCERT 10.1.1 PG. 163